Predicting Music Popularity on the SpotGenTrack Dataset: A Classification and Hypothesis-Testing Study

(Phase 1: Interim Report)

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***Abstract*—Predicting the popularity of a track before or shortly after its release is a long-standing problem in Music Information Retrieval (MIR) with direct commercial relevance for streaming platforms. This interim report describes the first phase of a term project built on the SpotGenTrack Popularity Dataset (SPD), which covers 101,939 tracks collected from the Spotify Web API. Two research questions are formulated: (i) a multi-class classification problem that predicts three popularity tiers of a track from audio features and metadata, and (ii) a hypothesis-testing problem that evaluates whether the musical mode (major vs. minor) is associated with a statistically significant difference in popularity. We present a brief literature review, a data quality and descriptive analysis of SPD, a naive majority-class baseline, and a justification for the chosen performance metrics (Macro-F1 and Balanced Accuracy).**

***Index Terms*—Music information retrieval, popularity prediction, SpotGenTrack, classification, hypothesis testing, feature engineering**

I. Introduction

The global streaming industry has transformed music discovery into a data-driven process in which a track's digital popularity score becomes both a marketing signal and an input to recommendation systems. Predicting this score from objective audio and textual features is one of the canonical tasks of Music Information Retrieval (MIR), often referred to as *Hit Song Science*. Although the overall popularity of a song is driven by factors that are hard to quantify (artist reputation, marketing, social media exposure), empirical work has shown that audio characteristics and metadata carry a non-trivial predictive signal.

Three peer-reviewed studies are particularly relevant here. Martín-Gutiérrez *et al.* [1] introduced SPD together with *Hit-MusicNet*, a multimodal end-to-end deep learning architecture that jointly processes audio, lyrics and metadata, with reported accuracy and recall around 83%. Their paper established SPD as a reproducible benchmark for popularity-prediction models. Khan *et al.* [2] looked at the effect of correlation-based filter feature selection on music popularity classification using classical machine learning algorithms on a large Spotify dataset; the reduced feature set preserved accuracy (95.14%) while lowering computational cost, which is one reason we include an explicit feature-importance analysis in our own modeling. Saragih [3] compared the classification and regression models on the Spotify audio features for Indonesian listeners. Tree-based models (Random Forest for classification, Extra Trees for regression) clearly outperformed linear baselines, but precision on highly popular tracks was substantially lower than on unpopular ones. That asymmetry is directly relevant for our choice of class-balanced evaluation metrics in Section IV.

What we draw from this short review is that multimodal features tend to help, that many Spotify descriptors are redundant and benefit from feature selection, and that the popularity label is imbalanced in a way that hurts naïve evaluation. Our project revisits SPD with these observations in mind and formulates two connected research questions (Section II).

II. Dataset and Research Questions

A. Dataset

SPD [4] is a public MIR dataset built on information collected from the Spotify Web API. It contains 101,939 tracks from 56,129 artists and 75,511 albums, covering the top 50 playlists across 26 countries. The dataset is distributed as several relational tables covering: *(a)* track-level Spotify audio features (danceability, energy, valence, acousticness, instrumentalness, liveness, speechiness, loudness, tempo, duration, key, mode, time signature), *(b)* track metadata (title, album, market availability), and *(c)* artist metadata (followers, artist popularity, genres). The continuous target variable is the Spotify *track popularity* score, an integer in $[0, 100]$ derived by Spotify from recent stream counts. The dataset also ships a fourth table of lyric-derived features extracted from Genius, but we restrict this project to audio and metadata signals and do not use the lyric table.

To support a classification problem, the popularity score is discretized into three tiers using tertile thresholds of the empirical distribution. The tertile choice is robust to the specific shape of the distribution and produces a roughly balanced three-class problem on SPD, where popularity is reported to

follow an approximately Gaussian shape with $\mu \approx 40$ and $\sigma \approx 17$ [1].

B. Research Questions

RQ1 (Machine Learning). *To what extent can the three-tier popularity class of a SpotGenTrack track be predicted from its audio and metadata features, and which feature groups contribute most to this prediction?* The target is the categorical variable $Y \in \{\text{Low}, \text{Medium}, \text{High}\}$ obtained from the tertile binning above. The predictor set X is the union of the Spotify audio descriptors and the track/artist metadata listed above.

RQ2 (Hypothesis Testing). *Is the mean popularity of tracks written in a major key significantly different from that of tracks written in a minor key?* The musical `mode` attribute provided by the Spotify API is a binary flag: 1 indicates a major key and 0 a minor key. Let P_1 denote the popularity of tracks with `mode`$= 1$ and P_0 that of tracks with `mode`$= 0$. We formulate

$$\mathcal{H}_0 : \ \mu_{P_1} = \mu_{P_0}, \qquad \mathcal{H}_1 : \ \mu_{P_1} \neq \mu_{P_0}, \tag{1}$$

with a two-tailed significance level $\alpha = 0.05$. The default test is the two-sample Student's t-test. Its two main assumptions (approximate normality of the group means and equal variances) will be checked in Phase 2 using a Q–Q plot and Levene's test; if either assumption is clearly violated, we will fall back to the non-parametric Mann–Whitney U test, which targets a difference in distribution location rather than means. The very large sample size also invokes the central limit theorem for the sampling distribution of the mean, which makes the t-test robust to moderate departures from normality.

The two research questions are connected by design. If RQ2 rejects $\mathcal{H}_0$, then the `mode` feature should carry measurable importance in the RQ1 classifier; if it does not, its low importance under feature attribution will act as a cross-check on the test's finding.

III. Data Quality and Descriptive Analysis

A. Data Profiling Pipeline and Findings

The quality checks are implemented in the `notebooks/01_data_exploration.ipynb` notebook of the replication package and cover five standard axes. *Completeness*: the `tracks` table (101,939 rows $\times$ 32 columns) has no missing values; the `artists` table has exactly one missing entry in the `name` column. No imputation is therefore required for the classification features. *Uniqueness*: all Spotify track identifiers are unique, but the key (`name`, `artists_id`, `duration_ms`) produces 2,438 duplicate rows, which correspond to the same recording released in multiple regional markets. These will be collapsed in Phase 2 before modeling. *Validity*: range checks on the Spotify audio features pass for all $[0, 1]$-bounded descriptors and for `tempo`; `loudness` is bounded at ≤ 0 dB but 42 rows sit marginally above zero (up to 2.72 dB), which is a known artefact of the Spotify loudness estimator — these will be clipped to 0 dB in Phase 2. *Consistency*: track-to-artist joins on `artists_id` succeed for the full cross product. *Distributional check*: the empirical popularity has $\mu = 39.78$, $\sigma = 16.79$, which is consistent with the $\mu \approx 40$ and $\sigma \approx 17$ reported by [1] and confirms that the downloaded copy matches the published dataset.

TABLE I
Descriptive statistics for the selected features on SpotGenTrack ($N = 101{,}939$).

Feature	Mean	Std. Dev.	Median	Min	Max
popularity	39.78	16.79	41.00	0	97
danceability	0.59	0.18	0.61	0.00	0.99
energy	0.59	0.26	0.63	0.00	1.00
valence	0.48	0.26	0.48	0.00	0.99
acousticness	0.35	0.33	0.24	0.00	1.00
instrumentalness	0.15	0.30	0.00	0.00	1.00
speechiness	0.13	0.20	0.05	0.00	0.97
liveness	0.20	0.18	0.12	0.00	1.00
loudness (dB)	−9.46	6.20	−7.60	−60	2.72
tempo (BPM)	118.36	30.22	118.07	0	244.04
duration (s)	246.77	190.43	216.89	1.16	5,505.83

Outliers in `tempo` and `duration_ms` will be winsorized at the 1st and 99th percentiles in Phase 2. Numerical features will be standardized (zero mean, unit variance) for the scale-sensitive models only (e.g. logistic regression), while tree ensembles will receive the raw scale.

B. Descriptive Statistics

Table I summarizes the distribution of the numerical features that feed both research questions. Figure 1 shows the empirical distribution of the target variable together with the tertile cut-points at $t_1 = 33$ and $t_2 = 48$ that split the data into the Low, Medium and High popularity classes used in RQ1 (class shares after the split: 32.1%, 34.2% and 33.7%, respectively, so the problem is essentially balanced). For RQ2, the `mode` attribute (major vs. minor key) provides the binary grouping variable whose effect on popularity will be tested.

IV. Evaluation Plan

A. Performance Metrics

Classification performance in Phase 2 will be reported on *Macro-F1* and *Balanced Accuracy*, both averaged across the three popularity tiers.

Macro-F1 is the unweighted mean of the per-class F1 scores and is the recommended metric for imbalanced multi-class problems because each class contributes equally regardless of its support; a classifier that only predicts the majority class therefore receives a low Macro-F1. Its weakness is that it conflates precision and recall into a single number and is less interpretable than raw accuracy.

Balanced Accuracy, defined as the arithmetic mean of per-class recalls, protects against the same pathology as Macro-F1 but is more sensitive to the true-positive rate than to precision, making it a useful complement: a model with high Macro-F1 but low Balanced Accuracy is sacrificing recall on a minority class, which is exactly the failure mode reported by Saragih [3]

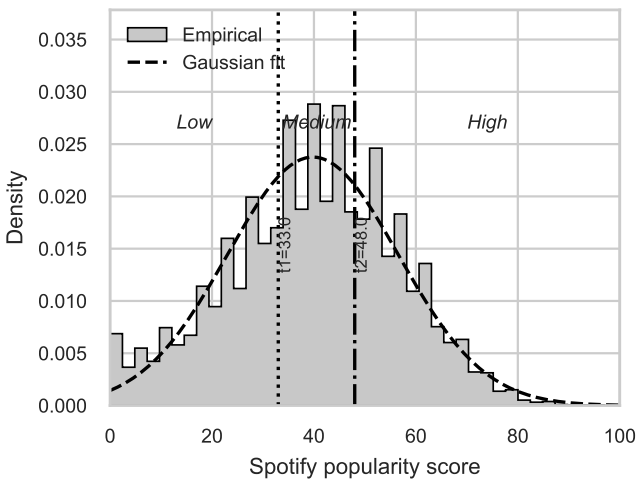

Fig. 1. Empirical distribution of the Spotify popularity score on SPD (histogram with kernel density overlay). Vertical dashed lines show the tertile cut-points that define the three classes Low, Medium and High used in RQ1. Different line styles are used so that the plot is readable in a black-and-white print.

TABLE II
PERFORMANCE OF THE TWO NAIVE BASELINES ON SPOTGENTRACK UNDER STRATIFIED 5-FOLD CROSS-VALIDATION (MEAN $\pm$ STD. DEV. ACROSS FOLDS).

Baseline	Macro-F1	Balanced Accuracy
Majority-class	0.170 ± 0.000	0.333 ± 0.000
Stratified random	0.337 ± 0.003	0.337 ± 0.003

for highly popular tracks. The two metrics together therefore cover precision–recall trade-offs across strata.

Both metrics will be reported with 95% confidence intervals obtained from stratified 5-fold cross-validation, so that differences between the naive baseline, the two ML models and the tuned model can later be compared with the paired statistical tests recommended by Demšar [5].

B. Naive Baseline

Two parameter-free reference classifiers are evaluated on the full SPD. The *majority-class classifier* always predicts the most frequent popularity tier (Medium, with a 34.2% share); the *stratified random classifier* draws predictions from the observed class prior. Neither requires any training, so both are trivially reproducible. Table II reports their Macro-F1 and Balanced Accuracy under stratified 5-fold cross-validation. The majority-class baseline collapses to Macro-F1 $= 0.17$ because only one of the three classes contributes a non-zero per-class F1; the stratified baseline reaches the expected $1/3 \approx 0.337$ on both metrics. Any model proposed in Phase 2 will have to beat these numbers by a statistically significant margin, following the comparison protocol of Demšar [5], before it can be considered useful.

V. SUMMARY AND NEXT STEPS

This interim report fixed the scope of the project on SpotGenTrack and set up the scaffolding that Phase 2 will build on. Two connected research questions were defined, one classification problem on three popularity tiers and one hypothesis test on the musical-mode (major vs. minor) attribute, and three peer-reviewed references from IEEE, MDPI and Taylor & Francis publishers were used to place the problem in context. The data-quality pipeline described in Section III covers the checks that are worth running on SPD before modeling, and Section IV explains why Macro-F1 and Balanced Accuracy are the right metrics here and how the majority-class baseline sets a lower bound of 0.17 on Macro-F1.

Phase 2 will complete the classification pipeline: feature engineering with correlation-filter feature selection in the spirit of [2], two learning algorithms (a regularized linear model and a gradient-boosted tree ensemble), hyper-parameter tuning, a statistical comparison against the baseline following Demšar [5], and a feature-importance analysis. The outcome of the feature-importance analysis will be cross-checked against the mode-based hypothesis test, so that the two research questions can speak to each other rather than being reported side by side.

AI Usage Statement. Generative AI tools were used for language editing and as an aid during the literature search, and also data extracting and visualising in coding implementation. All content was reviewed and verified by the author.

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